

flashjet

GPU-native jet clustering with the full merge history

Chirayu Gupta^{1,*} Sitian Qian² Alexandre De Moor¹

¹Vrije Universiteit Brussel (BE)

²Peking University (CN)

ML4Jets 2026 · Vienna · September 2026

* speaker

- 1 Jet clustering, and flashjet
- 2 Correctness
- 3 Speed
- 4 Downstream: the clustering tree as tagger input

Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not

1. Where the field is

- CMS has run GPUs in the High Level Trigger since Run 3
- $\sim 30\%$ of HLT reconstruction is offloaded $\rightarrow \sim 25\%$ less HLT time
- ported: pixel local reco · pixel tracks and vertices · calorimeter local reco

2. What is missing

Jet clustering is **still on the CPU**.

3. Why it stayed there

- sequential recombination is **IRC safe** — that is why it won
- but each merge **depends on the previous one**
- a serial data dependence is the one thing a GPU does badly

4. Why that costs us now

- reclustering **inside a training loop**
- scanning the **jet radius R**
- **substructure** computed on demand

every one of these pays a host $\leftrightarrow$ device round trip, once per call

What it does

- generalised- k_t clustering — anti- k_t , k_t , Cambridge–Aachen
- runs on the GPU, **many jets or events per launch**
- returns jets, the **full merge history**, and substructure **as tensors**

Why it can be done at all

- Cacciari–Salam nearest-neighbour lemma: the smallest d_{ij} in the *whole event* is always some particle's **geometric** nearest neighbour
 - so each particle only needs to track **its own** NN ...
 - ... and one merge invalidates only $O(1)$ of them
- ⇒ $O(N^2)$ **instead of** $O(N^3)$ — the serial chain stays, but each step becomes cheap and parallel

In the ML ecosystem

- a **PyTorch** call — tensors in, tensors out
- no host ↔ device copy in a training loop

Written in **Triton**: compiles and autotunes for whatever GPU it finds, so there is no separate CUDA build to maintain and results are reproducible once the tuning is cached.

One exponent, three algorithms

The generalised- k_t distance measure:

$$d_{ij} = \min(p_{Ti}^{2p}, p_{Tj}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$

$$d_{iB} = p_{Ti}^{2p}$$

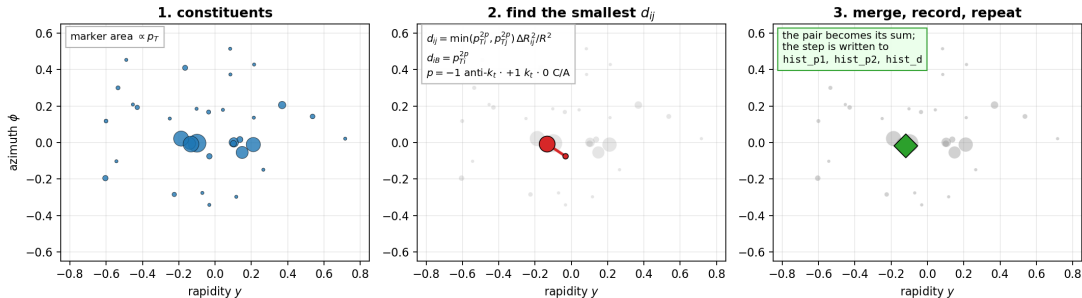
Merge the pair with the smallest d_{ij} ; if the smallest distance is a d_{iB} , call i a jet and remove it.

Only the exponent p changes:

p	algorithm	merges first	tree ordered by	used for
-1	anti- k_t	the hardest pairs	—	finding jets
0	Cambridge–Aachen	the closest pair	angle	grooming, Lund plane
+1	k_t	the softest pair	relative p_T	exclusive subjets

anti- k_t answers *which particles form a jet*. The other two answer *how that jet came apart* — and the **ordering is what the substructure reads**.

Sequential recombination — and what flashjet keeps from it



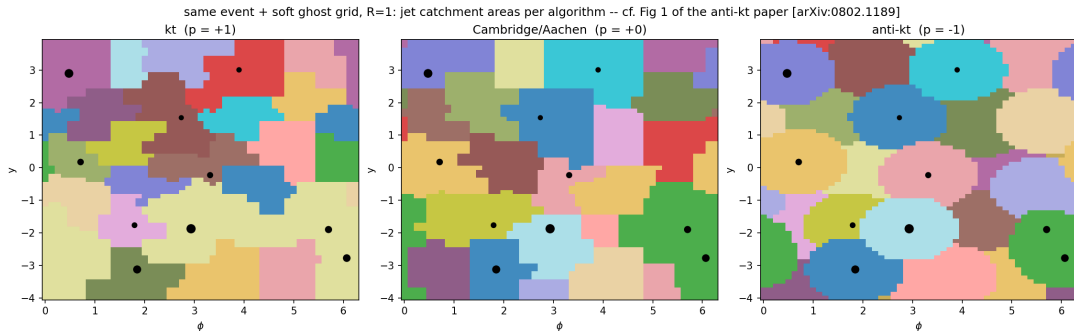
Two distances, computed for every pair:

- d_{ij} — *pair* distance, particle i to particle j
- d_{iB} — *beam* distance, particle i to the beam

Then repeat until nothing is left:

- ① find the smallest of all distances
- ② if it is a d_{ij} : **merge** i and j
- ③ if it is a d_{iB} : call i a **jet**, remove it

Both distances were defined on the previous slide; the exponent p there selects which algorithm this loop runs.



- jet catchment areas, all three algorithms
- clustered by **flashjet** itself

- k_t , C/A: **ragged**, fluctuating boundaries
 - anti- k_t : **circles of area πR^2** on each hard particle
- $\Rightarrow$ the algorithm's defining result, **reproduced without FastJet**

It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

field	shape	what it holds
jet_idx	(B, N)	which jet each particle ended up in
n_jets	$(B,)$	how many jets were found
hist_p1, hist_p2	(B, N)	which pair merged , at each step
hist_child	(B, N)	what they merged <i>into</i>
hist_d	(B, N)	the distance d_{ij} at that step

The tree is the product, not a by-product

The bottom three rows are the **complete binary merge tree**. Groomed mass, z_g , R_g , Lund coordinates and exclusive subjets are not separate algorithms — they are **coordinates on that one tree**. Compute it once, read them all off, **no second clustering pass**.

Three things are read off that tree — F1, F2, F3 — and then we look at it on a real jet.

What is implemented

Clustering

- generalised- k_t : anti- k_t , k_t , C/A
- merge history recorded in-kernel
- many jets clustered in one batched call
- scales from single jets to whole events

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

What it does. Undo the last merges of the sequence to expose exactly n subjects — or stop at a d_{cut} .

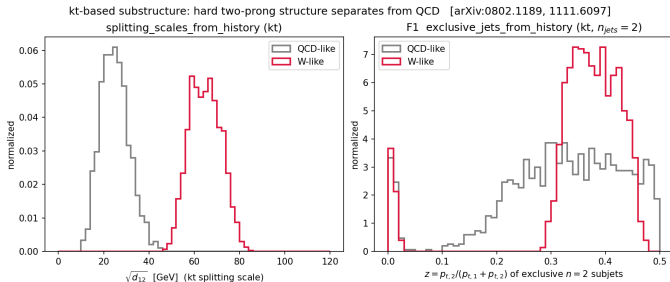
How. The history is already a binary tree. Walk back up it $n - 1$ steps; no reclustering.

Why k_t . k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

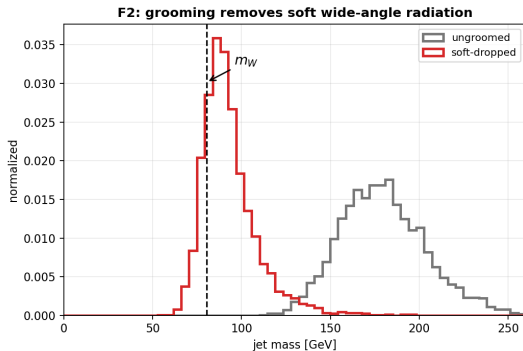
The splitting scales come straight from `hist_d`:

$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots$$

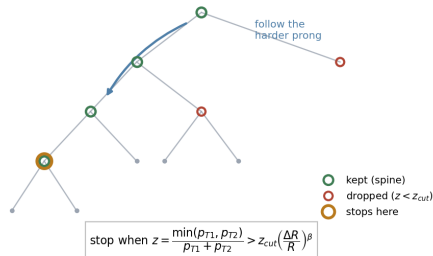
$$\text{with } d_{ij} = \min(p_{Ti}^2, p_{Tj}^2) \frac{\Delta R_{ij}^2}{R^2}$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjects — the W sits near $z \sim 0.4$, QCD spreads to small z .



the soft-drop walk on the C/A tree



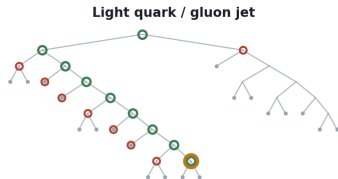
Stop when
$$\frac{\min(p_{T1}, p_{T2})}{p_{T1} + p_{T2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R} \right)^\beta$$

Walk down the C/A tree along the harder branch, discarding the softer prong until that condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

C/A merge trees of real jets — root at top, constituents at the bottom

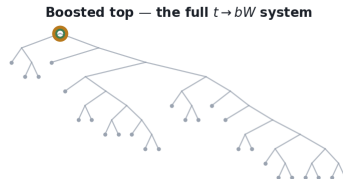
ATLAS Top Tagging Open Data · vertical = declustering depth · horizontal = angular ordering



a long spine: soft prong after soft prong
is stripped, and the mass collapses
 $n_{\text{drop}} = 8$ $m: 80 \rightarrow 1 \text{ GeV}$



the very first split is already balanced,
so nothing is groomed away
 $n_{\text{drop}} = 0$ $m: 80 \rightarrow 80 \text{ GeV}$



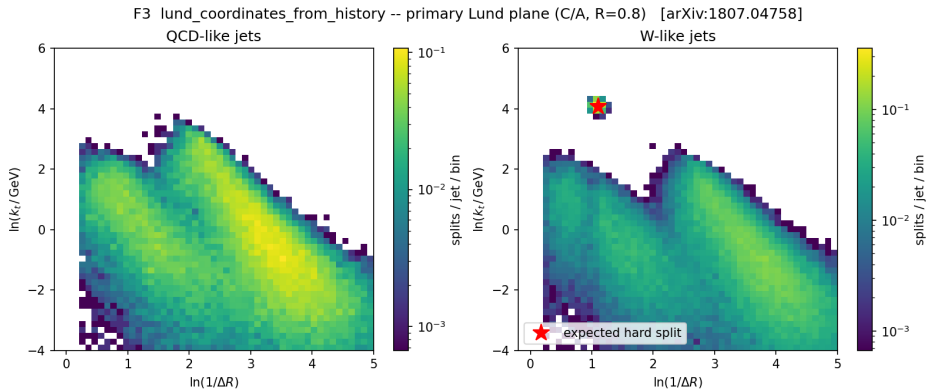
also stops at once, but on a wider split,
so the whole decay is kept
 $n_{\text{drop}} = 0$ $m: 170 \rightarrow 170 \text{ GeV}$

● soft-drop spine ● dropped prong ● grooming stops here ● constituent

A jet without a decay has nothing to stop on. The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

ATLAS Open Data, C/A with soft drop ($z_{\text{cut}}=0.1$, $\beta=0$). **The number of drops is itself a discriminant** — and it is read straight off the stored tree, at no extra cost.

A real decay stops at once. Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1} + p_{T2}}$, ΔR , $k_t = p_{T2} \Delta R$, and the plane coordinates $(\ln \frac{1}{\Delta R}, \ln k_t)$.

QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

How. One pass down the primary branch — each node is one point in the plane.

Check. The d -channel ties *exactly* to the splitting scales computed independently.

Correctness

Two kinds of validation

First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

Input A — toy fat jets — QCD-like and W-like fat jets

Input B — toy parton shower — leading-log parton shower

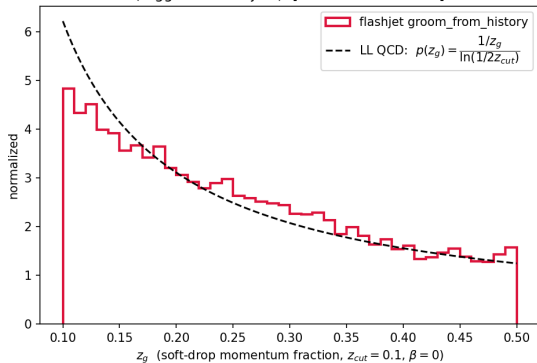
Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

Input C — ATLAS Open Data — large- R jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

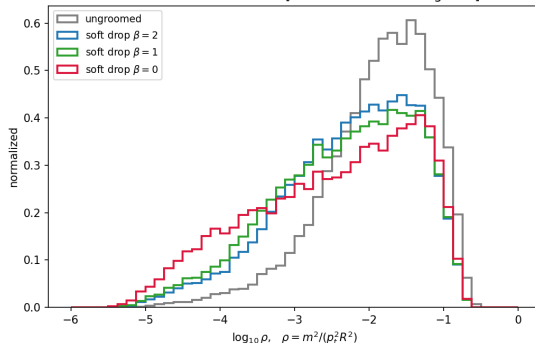
groomed splitting fraction vs the analytic prediction
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

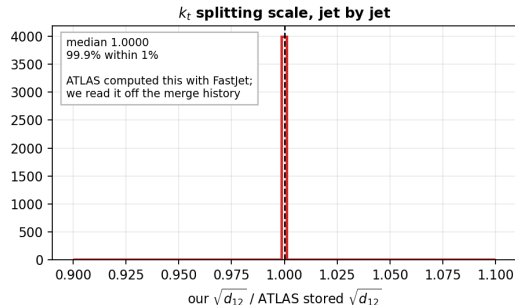
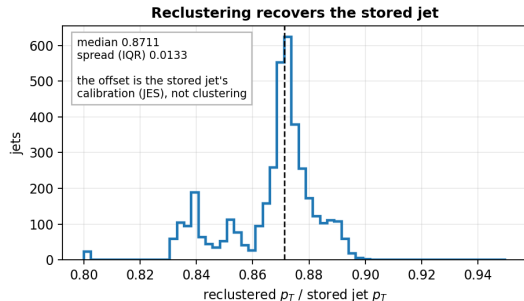
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller β) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in β : larger β grooms less, so the mass peak moves up.

Jet-by-jet closure against ATLAS-stored quantities



Left. Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

Right. The k_t splitting scale $\sqrt{d_{12}}$ against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

3 algorithms $\times$ 5 radii $\times$ 5 multiplicity bins $\times$ 2 samples, 20 000 jets per bin, on ATLAS Open Data.

check	result
number of jets found	100.000 % at every one of 150 points
leading-jet p_T within 10^{-4}	1.0000 at 148/150 points
median relative difference	$\sim 7 \times 10^{-8}$ (float32 precision)

The grid is the point

A single working point proves little. Agreement holds across **all three algorithms**, **every radius** from 0.4 to 1.2, and **24 to 123** constituents per jet.

Two independent references

The same events are clustered by **FastJet** for the jets, and every substructure decoder is separately pinned against an **independent NumPy tree-walk**.

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the ~ 600 clusters per event ourselves, and compare:

n_{jets} vs ATLAS	100.0 % match
mean jets/event	10.96 vs 10.96

This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is $O(N^2)$.

Speed

Data — ATLAS Open Data, freely presentable

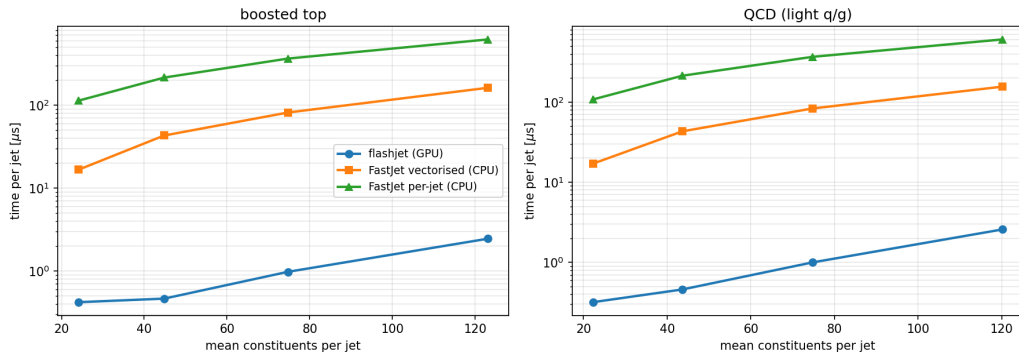
sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

Large- R jets with calorimeter-cluster constituents. The grid is **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins**, 20 000 jets per bin.

Method

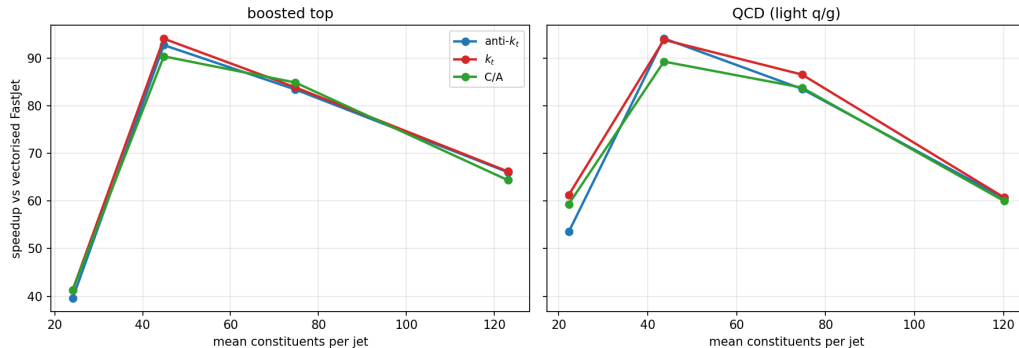
- 1 extract constituents once, save;
- 2 cluster on GPU, time it, save the *same* events;
- 3 cluster those saved events with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

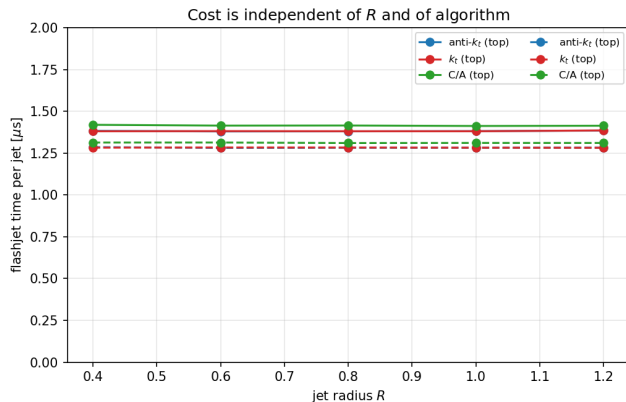
ATLAS Top Tagging Open Data — anti- k_t $R = 1.0$ 

Log scale — each gridline is $10\times$. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **0.35–2.7 μ s per jet** against **17–155 μ s** for vectorised FastJet, and **110–620 μ s per jet** one at a time. Benchmarked on an NVIDIA V100-class GPU.

Speedup vs multiplicity — $R = 1.0$ (all points 100% n_{jets} agreement)



All three algorithms, both samples, $R = 1.0$. **39–99 $\times$** over vectorised FastJet across the grid (median **64 $\times$**), and every point has **100 %** n_{jets} agreement. Busier jets $\rightarrow$ bigger win — the regime that matters for boosted-object physics.



Across $R = 0.4 \rightarrow 1.2$: a **0.3 % spread** over a $3\times$ range in R .

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- k_t , k_t and C/A agree to within $\sim 4\%$ at every multiplicity.

k_t and C/A cost the same as anti- k_t .

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E) ; CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" "kt" "cambridge"
p	float	generalised- k_t exponent (overrides algorithm)

out.n_jets	# (B,)	jets found
out.jets_p4(p4)	# (B,J,4)	jet four-momenta
out.splitting_scales()	#	$\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ...
out.exclusive_jets(n_jets=3)	# F1	exclusive k_t subjects
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)	# F2	soft drop / mMDT
out.lund_coordinates(p4, R)	# F3	primary Lund plane

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

Downstream: the clustering tree as tagger input

Does the tree help a tagger? — setup

The question. Gouskos & Maier [PLuM, 2605.26821] report that explicit Lund-plane splittings help a transformer tagger. We test that on a **full 10-class** training.

Model. Particle Transformer [2202.03772] — 8 encoder + 2 class-attention blocks, 8 heads, embed 128. **2.144 M** params baseline, **2.194 M** with the feature block (+0.051 M).

Data. JetClass, 10 classes, **~20.05 M** test jets (**~2.0 M/class**), 128 particles/jet. All arms read **identical** shards; features computed live.

Ranger, lr 10^{-3} , batch 512, 10^6 iterations, checkpoint every 20k. Uncertainties: bootstrap of the test set, 16–84th percentile.

The feature arms

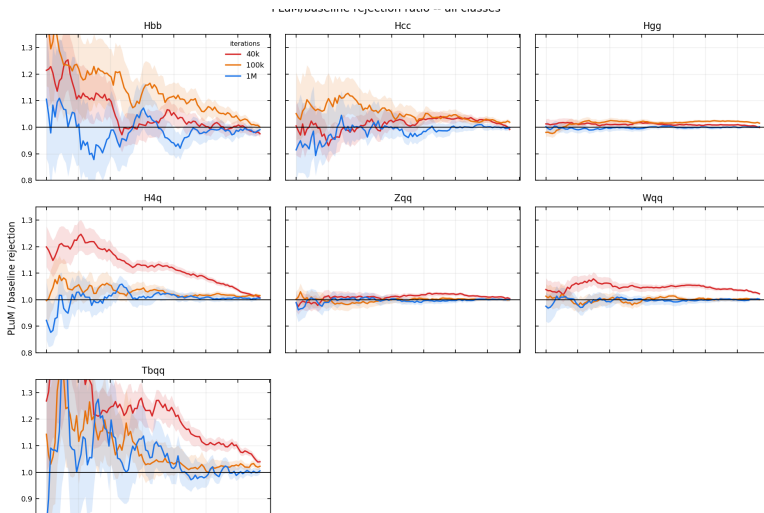
- **PLuM** — 48 k_t -ordered Lund splittings, $[\ln k_t, \ln 1/\Delta R, \ln z]$, embedded and **concatenated** as extra tokens (176 vs 128)
- **subject** — n -body subject mass, p_T fraction, n_{const}
- **CA5** — per-particle C/A branch-point features

Cost (wall clock per step)

PLuM	1.59×
subject	1.42×
CA5	1.22×

One seed per arm — see the caveat on the next slide.

Does the tree help a tagger? — result



PLuM/baseline rejection ratio; unpaired-bootstrap bands.

Test accuracy ~ 20.05 M jets

	40k	100k	1M
baseline	83.431	84.632	86.212
PLuM	83.428	84.697	86.206
Δ	-0.003	+0.065	-0.006

What we see

An early gain in every class, gone by 1 M. Wqq has the *fewest* splittings (21.6/jet) yet gains at $\sim 9\sigma$: **not confined to complex or *b*-jets**. At convergence: **null-or-worse**.

What we do *not* claim

- **No mechanism** — *b*-specificity excluded.
- **One seed per arm** — a second seed's spread exceeded the effect.
- Their Hbb reproduced at 100k, **not** at convergence.

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjects, soft drop, Lund coordinates — comes free as array reads.
- 2 **Gives the same jets as FastJet.** 150/150 grid points at 100 % n_{jets} agreement, across anti- k_t/k_t /C-A, $R = 0.4\text{--}1.2$ and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at ~ 600 clusters per event.
- 4 **39–99 $\times$ faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

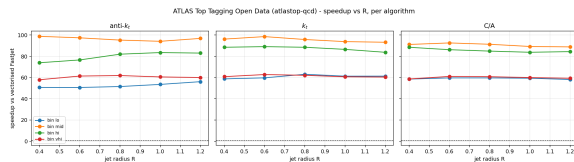
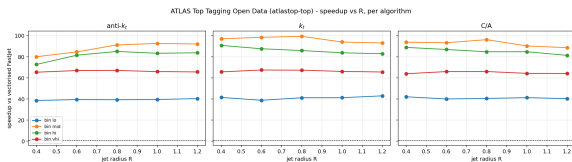
Thank you

Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

Backup

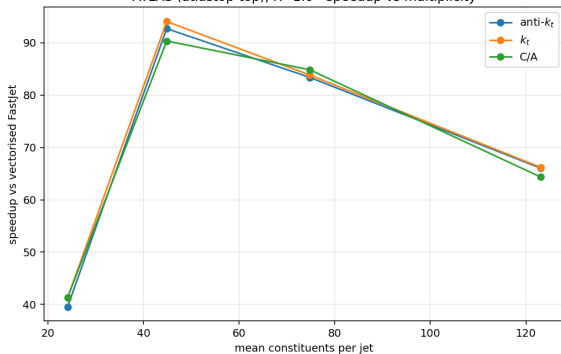


boosted top

Flat in R for all three algorithms; ordering is by multiplicity bin.

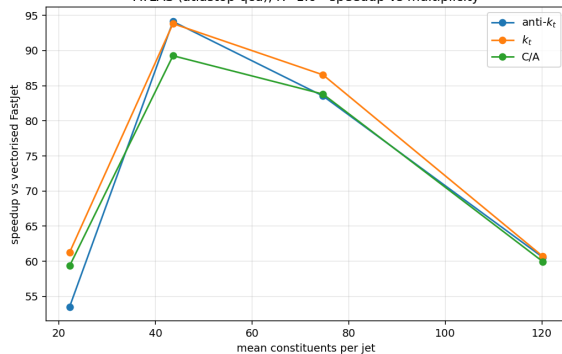
QCD (light q/g)

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light q/g)

Backup — the numbers

sample	alg	$\langle n \rangle$	flashjet [$\mu\text{s}/\text{jet}$]	FastJet vec. [$\mu\text{s}/\text{jet}$]	speedup
top	anti- k_t	24.1	0.427	16.5	39×
top	anti- k_t	44.8	0.464	43.0	93×
top	C/A	74.7	1.000	84.8	85×
top	C/A	123.1	2.565	165.1	64×
qcd	anti- k_t	22.2	0.346	17.5	51×
qcd	anti- k_t	43.6	0.463	45.7	99×
qcd	anti- k_t	74.7	1.215	89.8	74×
qcd	anti- k_t	120.2	2.673	154.9	58×

39–99× over vectorised FastJet (median 64×), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — where the time goes

Nsight Systems — share of GPU time

kernel	% GPU	avg/call
_cluster_large_kernel	97.6	5.47 ms
_decode_kernel	1.5	82 μ s
torch glue	< 1	—

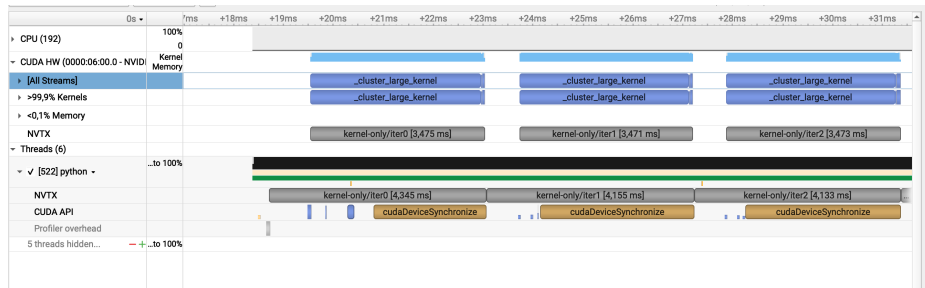
No hidden overhead — essentially all the time is real clustering work, and the memory row of the timeline is empty.

Nsight Compute — hardware counters

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

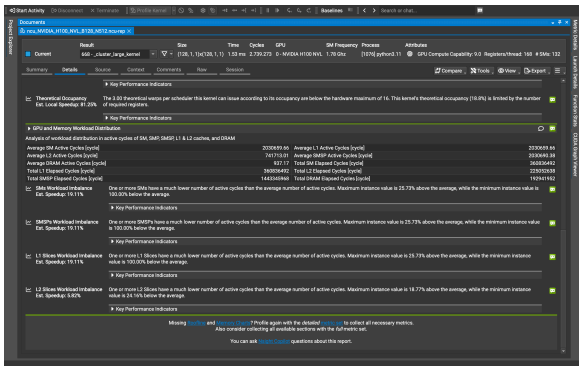
Not memory- or compute-bound — the GPU is simply **not filled**.

Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

Backup — the profiler names the bottleneck



The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81%

Achieved occupancy 67%

SM workload imbalance 19%

"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."

Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

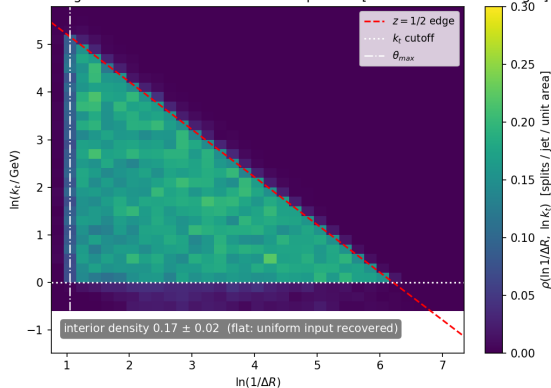
Backup — occupancy detail

Occupancy		
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.		
Theoretical Occupancy (%)	18.75	Block Limit Registers [block]
Theoretical Active Warps per SM [warp]	16	Block Limit Shared Mem [block]
Achieved Occupancy (%)	6.25	Block Limit Warps [block]
Achieved Active Warps per SM [warp]	16	Block Limit SM [block]
Cluster Occupancy (%)	0	Block Limit Registers [block]
Max Active Clusters [cluster]	8	Max Cluster Size [block]
Overall GPU Occupancy (%)	0	
Achieved Occupancy Ext. Local Speedup: 68.67%		
The difference between calculated theoretical (18.75%) and measured achieved occupancy (6.25%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the Occupancy Guide for more details on optimizing occupancy.		
Key Performance Indicators		
Theoretical Occupancy Ext. Local Speedup: 81.25%		
The 3.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 16. This kernel's theoretical occupancy (18.75%) is limited by the number of required registers.		
Key Performance Indicators		

Details		
Warning: Data collection happened without GPU frequencies being set by the profiler. Some results may be inaccurate.		
GPU Speed (0.1) Throughput		
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-unit of Compute and Memory to clearly identify the highest constraint.		
Compute (SM) Throughput (%)	15.40	Duration [ms]
Memory Throughput (%)	17.25	Elapsed Cycles [cycle]
L1TEX Cache Throughput (%)	33.23	SM Active Cycles [cycle]
L2 Cache Throughput (%)	0.82	SM Frequency [GHz]
DRAM Throughput (%)	0.02	DRAM Frequency [GHz]
Small Grid - This kernel grid is too small to fill the available resources on this device, resulting in only 6.32 full warps across all SMs. Look at the Launch Statistics for more details.		
Key Performance Indicators		
Launch Statistics		
Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.		
Grid Size	128	Function Cache Configuration
Cluster Size	0	Registers Per Thread [register/thread]
Cluster Scheduling Policy	Polygraph	Static Shared Memory Per Block [byte/block]
Block Size	128	Dynamic Shared Memory Per Block [byte/block]
Threads [thread]	16384	Driver Shared Memory Per Block [byte/block]
Waves per SM	0.32	Shared Memory Configuration Size [byte]
Warp Queue Context	0	Warp Size
# SMs (SM)	132	#TPCs
Enabled TPCs	all	
Small Grid - This kernel grid is configured to execute only 128 blocks, which is less than the 132 multiprocessors used. This can underutilize some multiprocessors. If you do not intend to execute this kernel concurrently with other workloads, consider reducing the block size to have at least one block per multiprocessor or increase the size of the grid to fully utilize the available hardware resources. See the Launch Statistics description for more details on launch configurations.		
Key Performance Indicators		
Occupancy		
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.		
Theoretical Occupancy (%)	18.75	Block Limit Registers [block]

Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

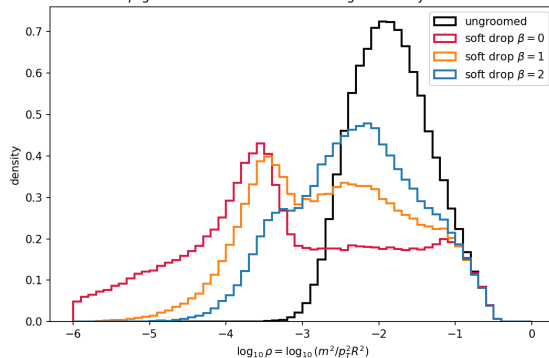
primary Lund plane of the toy shower (C/A, R=0.4):
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the d -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop β -ordering on REAL QCD jets (164292 jets, $z_{cut} = 0.1$)
smaller β grooms harder — same ordering as the toy-shower closure



The soft-drop β -family orders as predicted: larger β grooms less.

Backup — Input A: the toy fat jets

Purpose. QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like:** one hard collinear core (92 % of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8 %, $\sigma = 0.30$)
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV, $z \in [0.30, 0.45]$, random orientation, + 6 % soft contamination

Each “spray” is n massive pions with exponential p_T fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. `flashjet` has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep $v \leq \ln(p_{T0}/2) - u$ (i.e. $z \leq \frac{1}{2}$), $k_t > k_t^{\min}$

Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the $1/z$ form of z_g , the β -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

Jet areas use the anti- k_t -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers; fixed seed, fully re-runnable.

Backup — the validation ladder

rung	input	reference	result
unit tests	—	independent NumPy tree-walks	129 pass, incl. GPU paths
anti- k_t jet areas	toy B	the anti- k_t paper's own construction	rigid cones, $\rightarrow \pi R^2$
k_t substructure	toy A	known 2-prong vs QCD truth	clean separation
soft-drop z_g	toy B	leading-log $1/z$ prediction	on the curve
groomed mass vs β	toy B	Soft Drop Figs. 3–4	ordering reproduced
Lund plane	toy A	primary-Lund construction	hard split where predicted
jet-by-jet kinematics	real C	FastJet, same events	150/150 points, 100 % n_{jets}
full-event clustering	real C	the experiment's own jets	100 % match, ~ 600 clusters/ev

Each rung is closed against something independent — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

- H. Qu, C. Li, S. Qian, *Particle Transformer for Jet Tagging*, arXiv:**2202.03772**. (also the JetClass dataset; Zenodo 10.5281/zenodo.6619768)
- L. Gouskos, B. Maier, *Particle-Lund Multimodality in Jet Taggers*, arXiv:**2605.26821**.
- H. Qu, L. Gouskos, *Jet tagging via particle clouds* (ParticleNet), Phys. Rev. D **101** (2020) 056019, arXiv:1902.08570.
- F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP **12** (2018) 064, arXiv:1807.04758.
- F. A. Dreyer, H. Qu, *Jet tagging in the Lund plane with graph networks*, JHEP **03** (2021) 052, arXiv:2012.08526.
- A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP **05** (2014) 146, arXiv:1402.2657.
- M. Cacciari, G. P. Salam, G. Soyez, *The anti- k_t jet clustering algorithm*, JHEP **04** (2008) 063, arXiv:0802.1189; *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896, arXiv:1111.6097.
- S. Catani, Yu. L. Dokshitzer, M. H. Seymour, B. R. Webber, Nucl. Phys. B **406** (1993) 187 (k_t); Yu. L. Dokshitzer *et al.*, hep-ph/9707323 and M. Wobisch, T. Wengler, hep-ph/9907280 (Cambridge/Aachen).
- J. de Favereau *et al.*, *DELPHES 3*, JHEP **02** (2014) 057, arXiv:1307.6346.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.